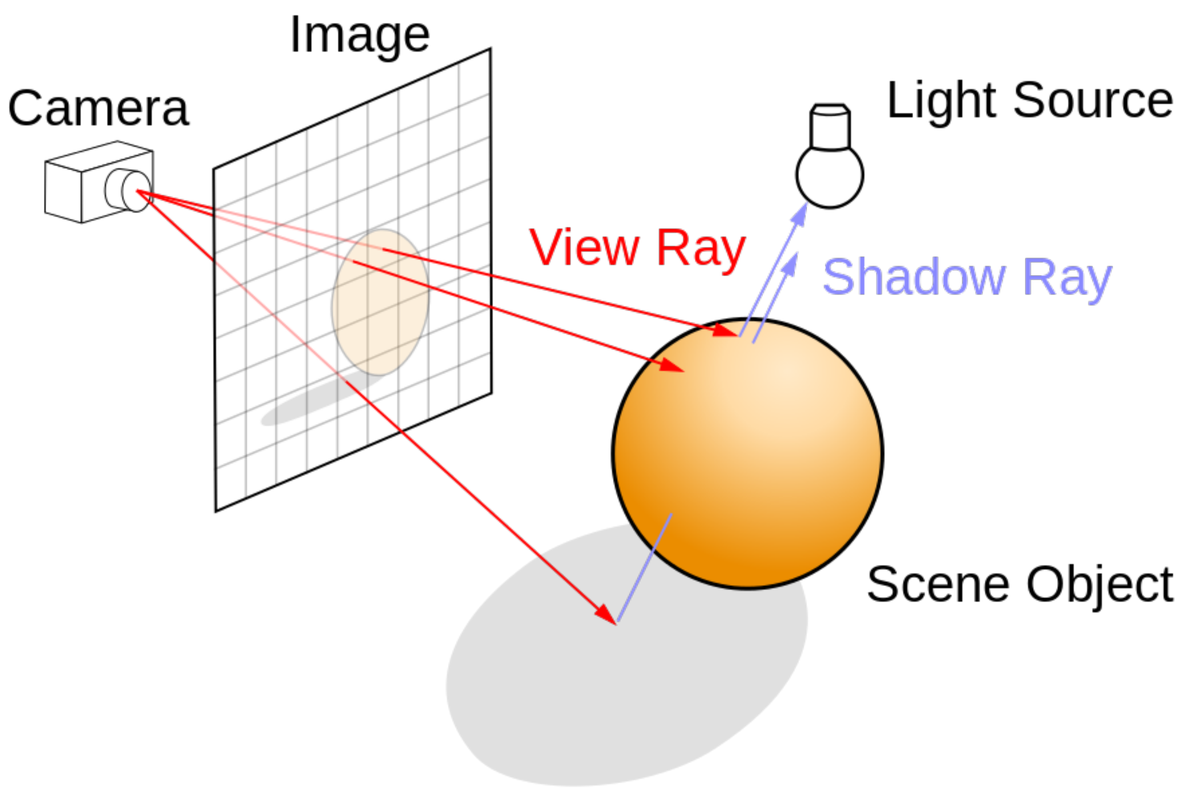
Rasterization

Acknowledgements: Daniele Panozzo

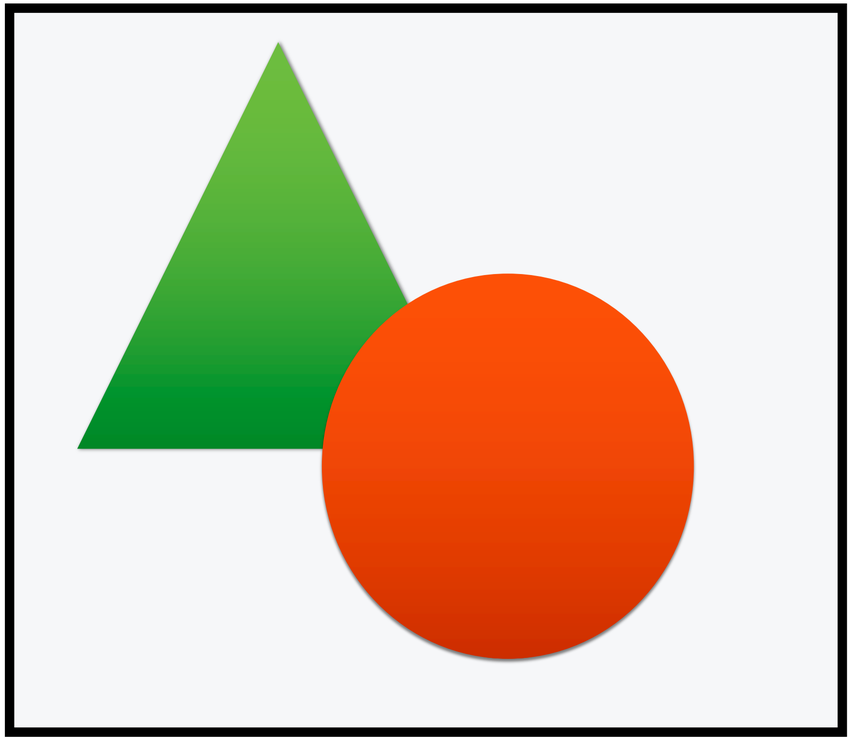
Two major approaches

Per-pixel - “Raytracing”



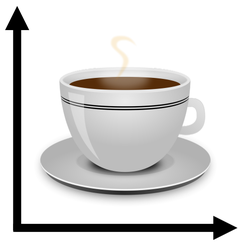
By Henrik - Own work, GFDL, https://commons.wikimedia.org/w/index.php?curid=3869326

Per-object - “Rasterization”

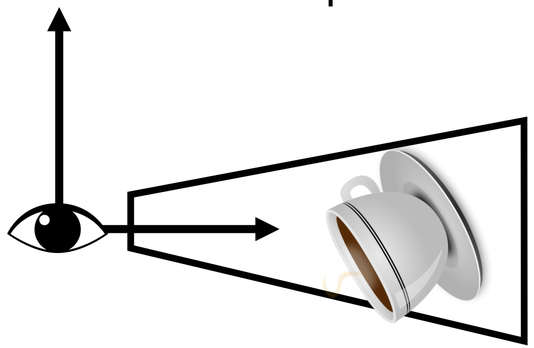


Recap: Viewing Transformation

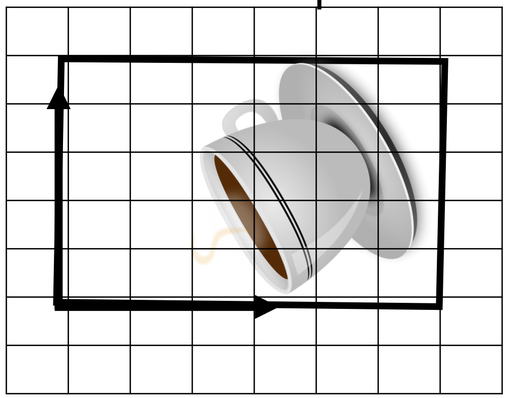
object space



camera space



screen space

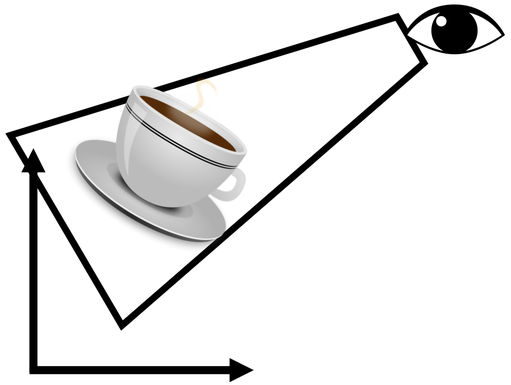


model

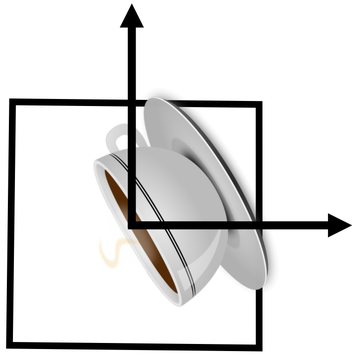
camera

projection

viewport



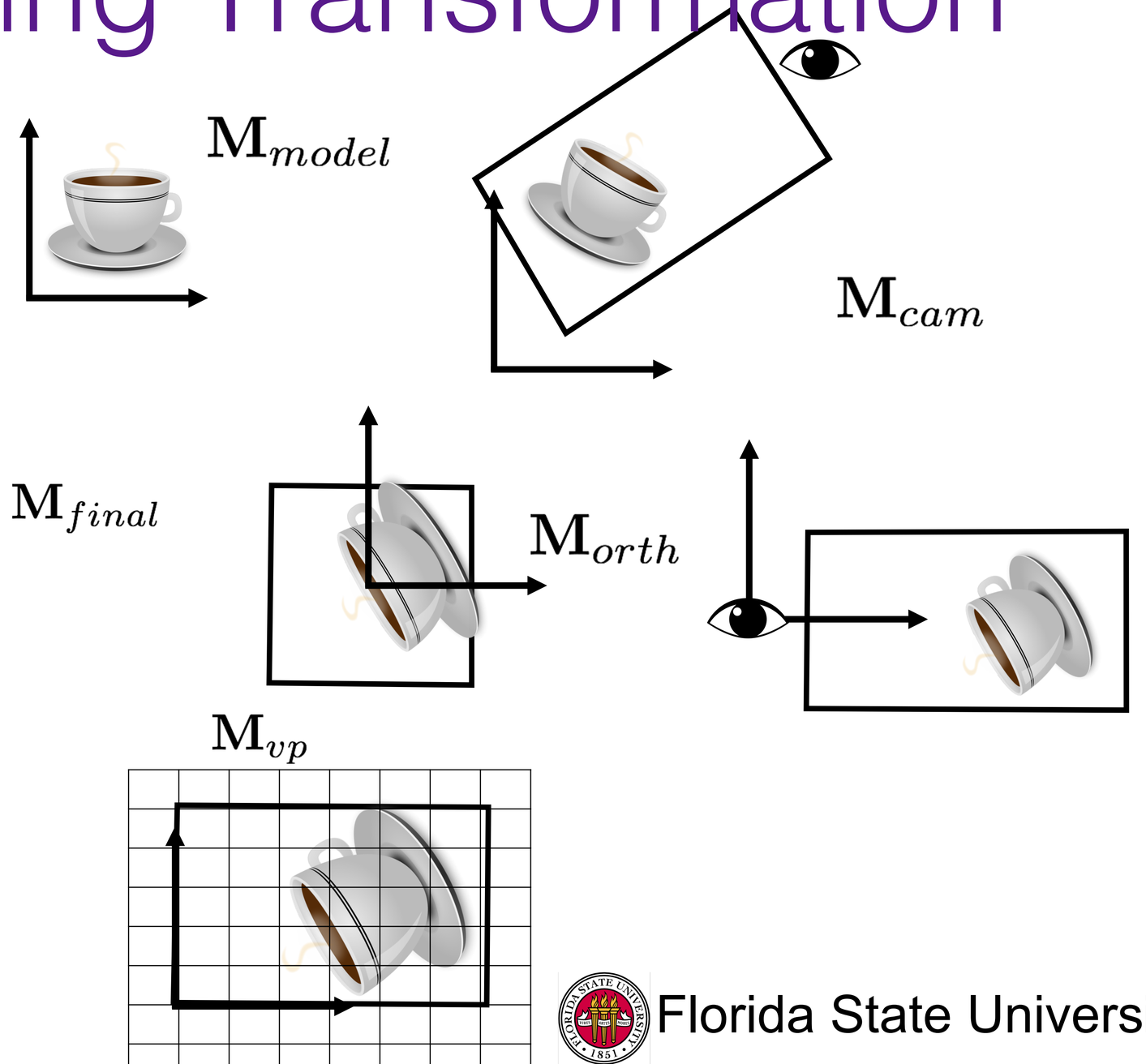
world space

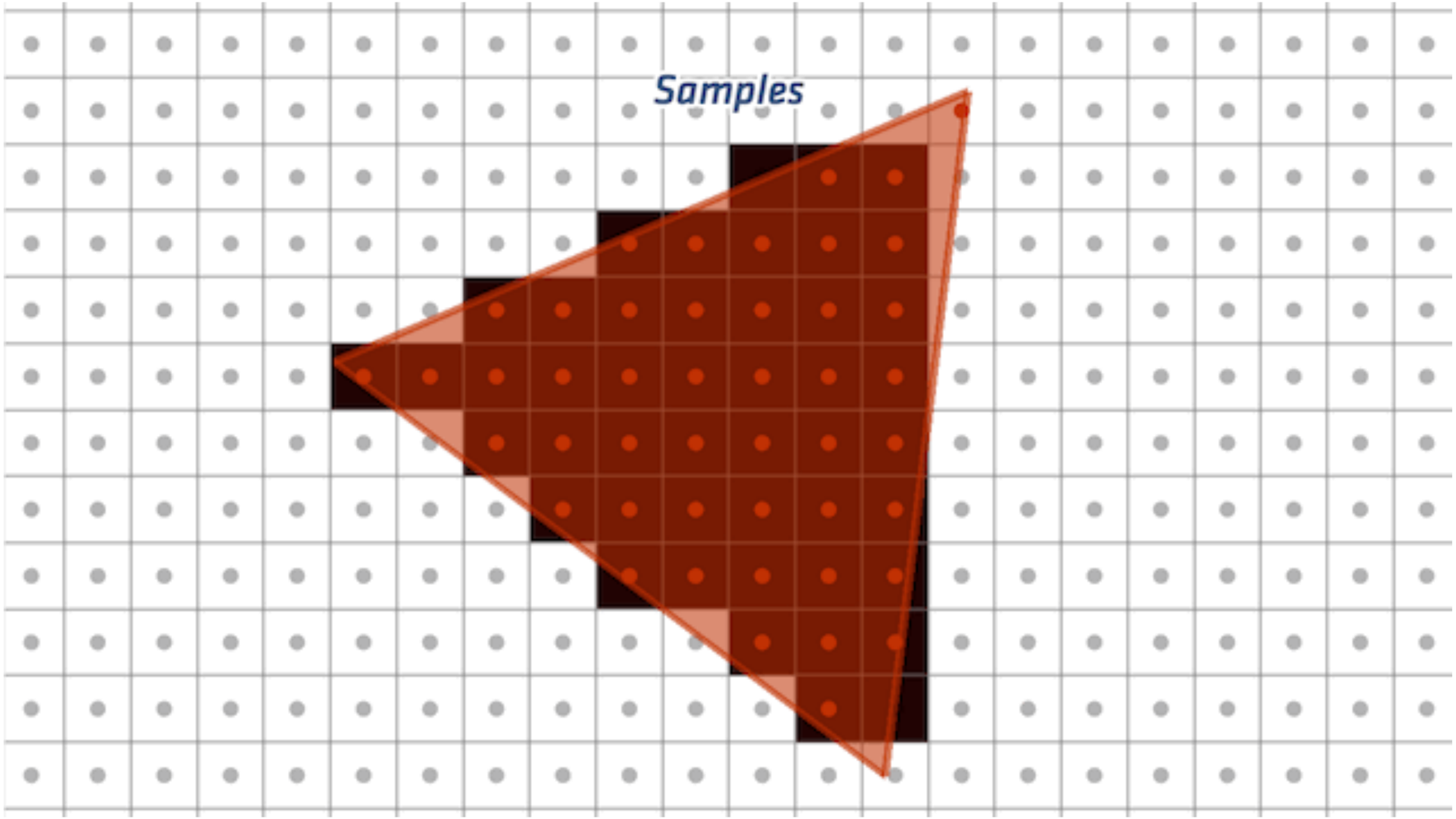


canonical
view volume

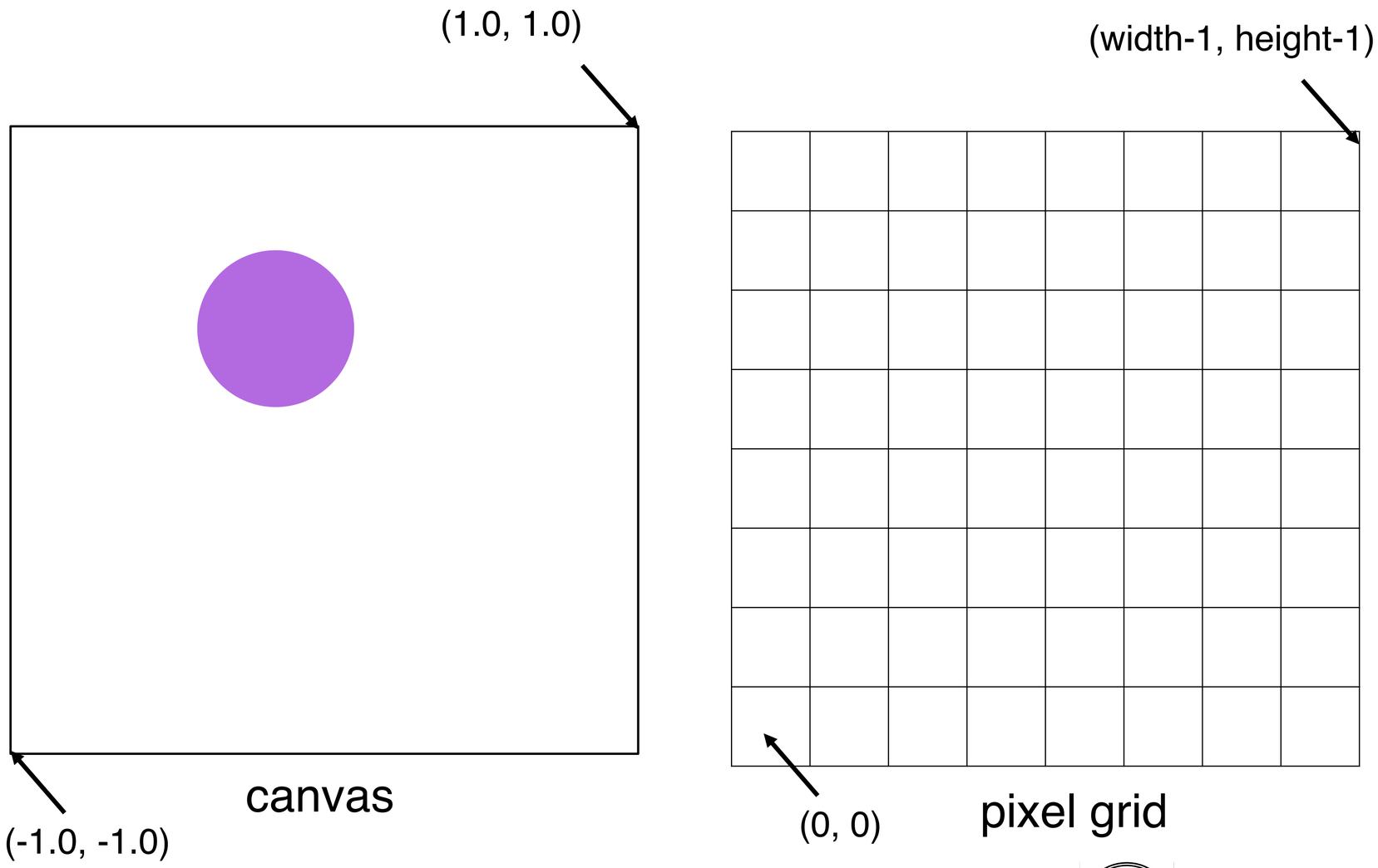
Recap: Viewing Transformation

- Construct Viewport Matrix $\mathbf{M}_{vp}$
- Construct Projection Matrix $\mathbf{M}_{orth}$
- Construct Camera Matrix $\mathbf{M}_{cam}$
- $\mathbf{M} = \mathbf{M}_{vp}\mathbf{M}_{orth}\mathbf{M}_{cam}$
- For each model
  - Construct Model Matrix $\mathbf{M}_{model}$
  - $\mathbf{M}_{final} = \mathbf{M}\mathbf{M}_{model}$
  - For every point **p** in each primitive of the model
    - $\mathbf{p}_{final} = \mathbf{M}_{final}\mathbf{p}$
  - Rasterize the model

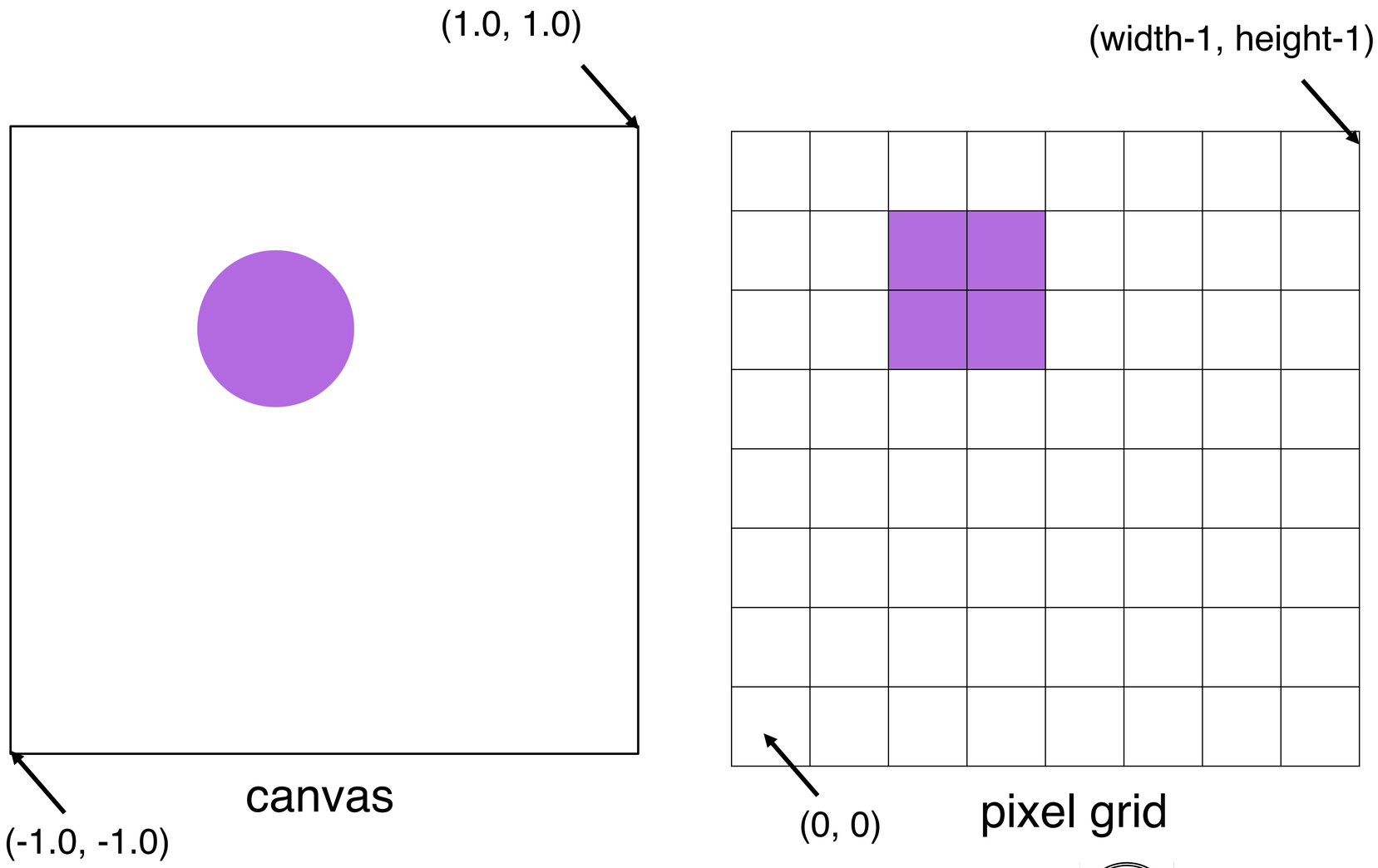



Samples

2D Canvas

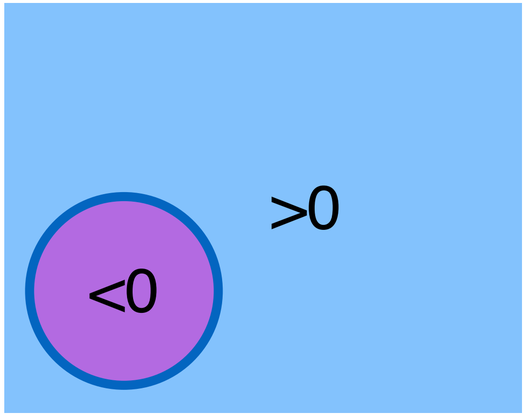


2D Canvas



Implicit Geometry Representation

- Define a curve as zero set of 2D implicit function
  - $F(x,y) = 0 \rightarrow$ on curve
  - $F(x,y) < 0 \rightarrow$ inside curve
  - $F(x,y) > 0 \rightarrow$ outside curve



- Example: Circle with center (c_x, c_y) and radius r

$$F(x,y) = (x - c_x)^2 + (y - c_y)^2 - r^2$$

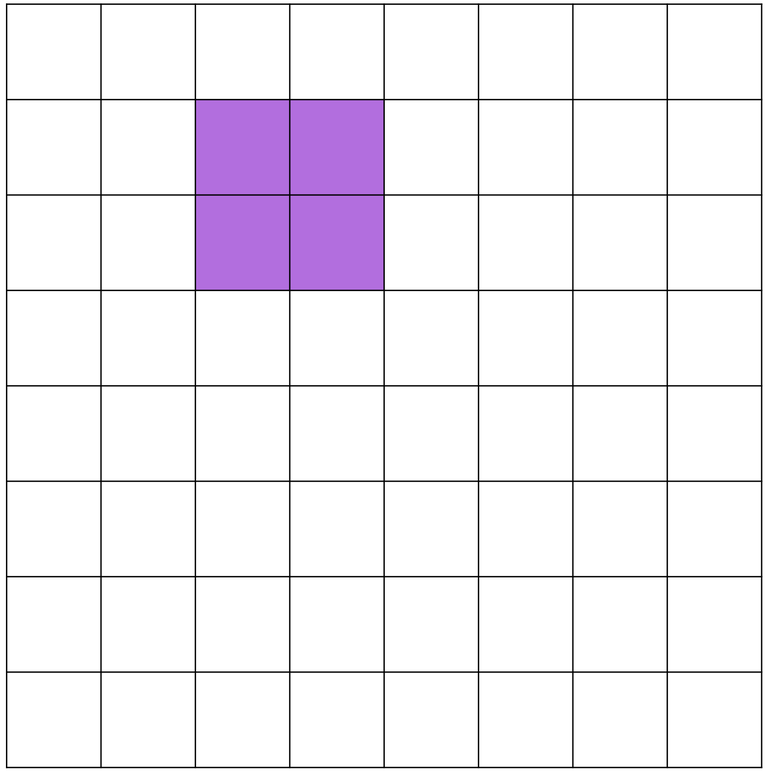
Implicit Rasterization

```
for all pixels (i,j)

        (x,y) = map_to_canvas (i,j)

       if F(x,y) < 0

        set_pixel (i,j, color)
```



Implicit Geometry Representation

- Example: Triangle

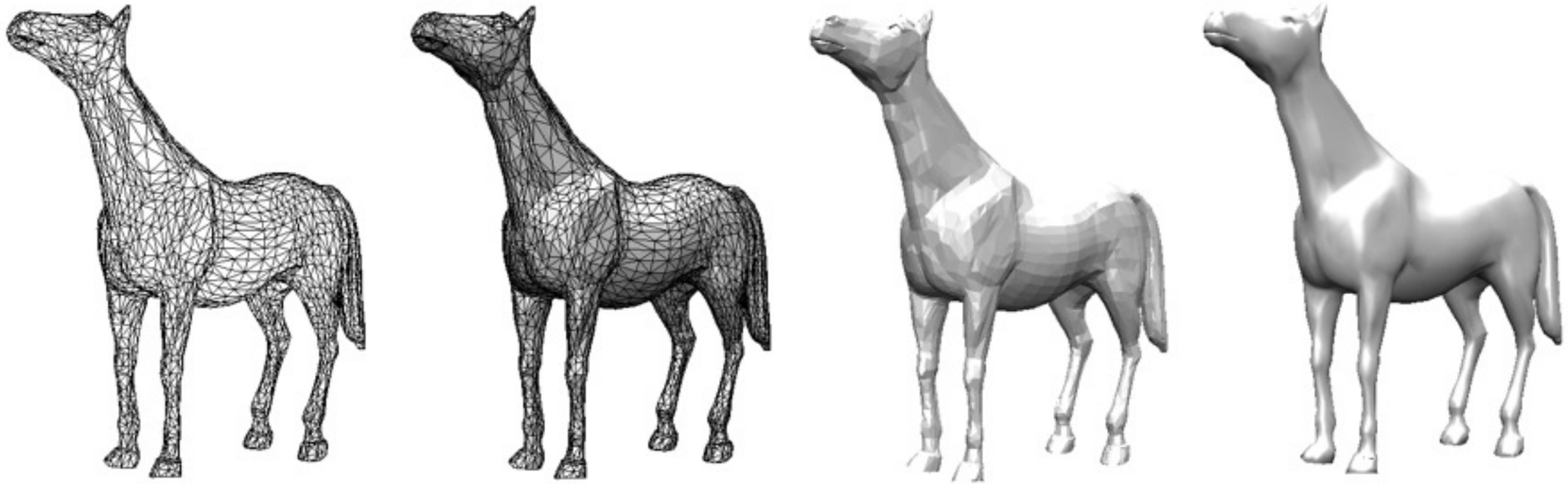
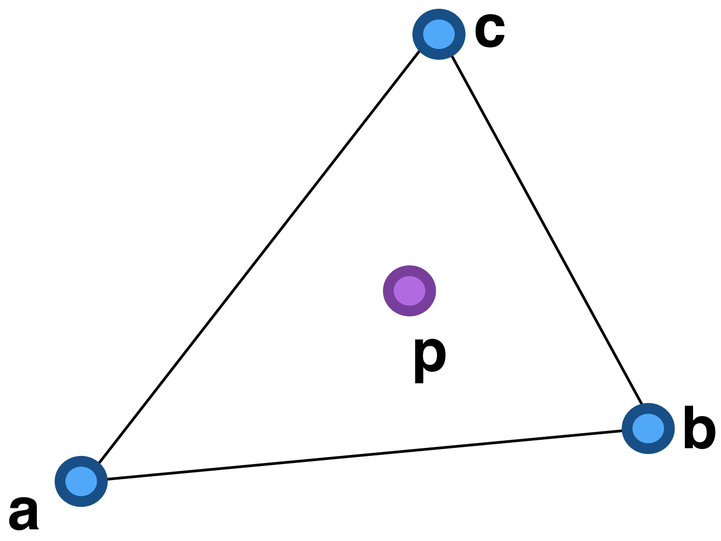


Image Copyright: Andrea Tagliasacchi

Barycentric Interpolation

- Barycentric coordinates:
  - $\mathbf{p} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}$ with $\alpha + \beta + \gamma = 1$



Barycentric Interpolation

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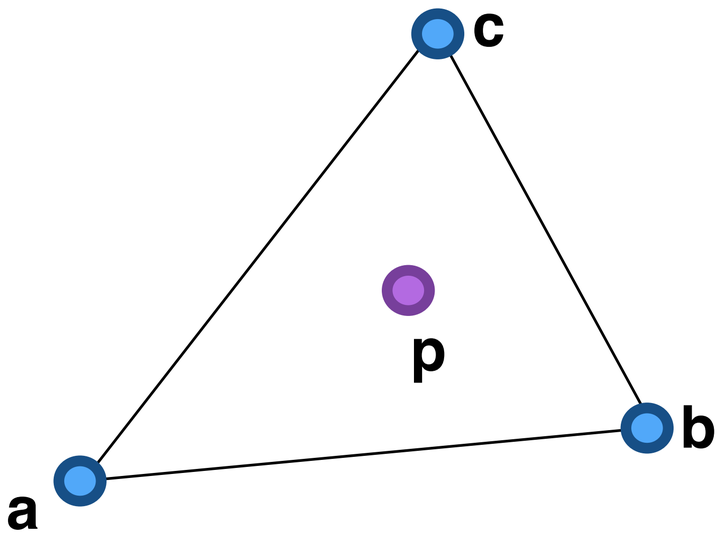
  - Unique for non-collinear **a**,**b**,**c**



$$\begin{bmatrix} \mathbf{a}_x & \mathbf{b}_x & \mathbf{c}_x \\ \mathbf{a}_y & \mathbf{b}_y & \mathbf{c}_y \\ 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} \mathbf{p}_x \\ \mathbf{p}_y \\ 1 \end{bmatrix}$$

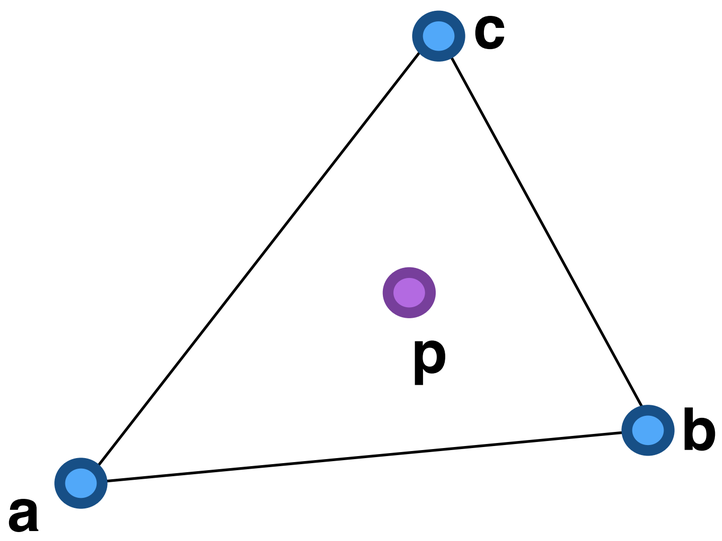
Barycentric Interpolation

- Barycentric coordinates:
  - $\mathbf{p} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}$ with $\alpha + \beta + \gamma = 1$
  - Unique for non-collinear **a**,**b**,**c**
  - Ratio of triangle areas

$$\alpha(\mathbf{p}) = \frac{\text{area}(\mathbf{p}, \mathbf{b}, \mathbf{c})}{\text{area}(\mathbf{a}, \mathbf{b}, \mathbf{c})}$$

$$\beta(\mathbf{p}) = \frac{\text{area}(\mathbf{p}, \mathbf{c}, \mathbf{a})}{\text{area}(\mathbf{a}, \mathbf{b}, \mathbf{c})}$$

$$\gamma(\mathbf{p}) = \frac{\text{area}(\mathbf{p}, \mathbf{a}, \mathbf{b})}{\text{area}(\mathbf{a}, \mathbf{b}, \mathbf{c})}$$



Barycentric Interpolation

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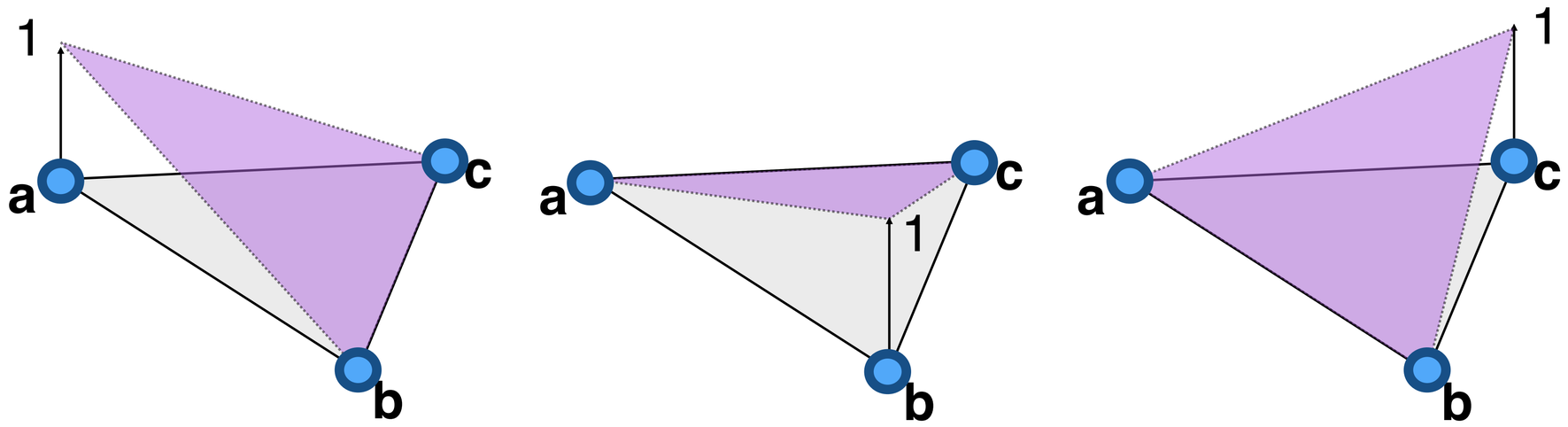
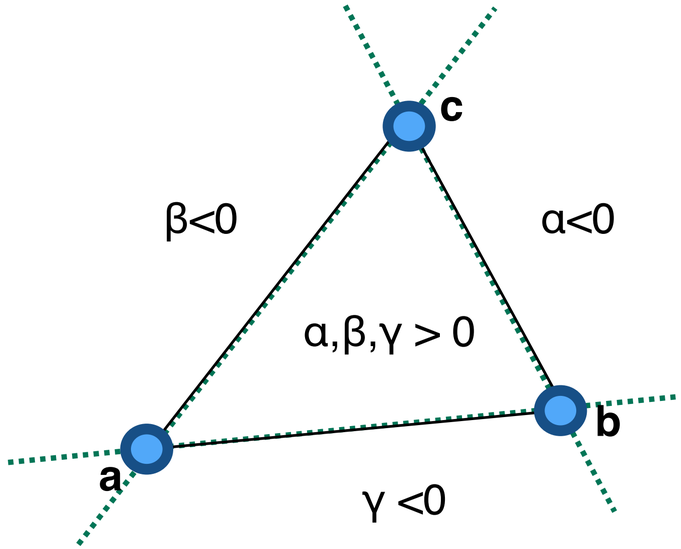
  - Unique for non-collinear **a**,**b**,**c**
  - Ratio of triangle areas
  - $\alpha(\mathbf{p})$, $\beta(\mathbf{p})$, $\gamma(\mathbf{p})$ are linear functions



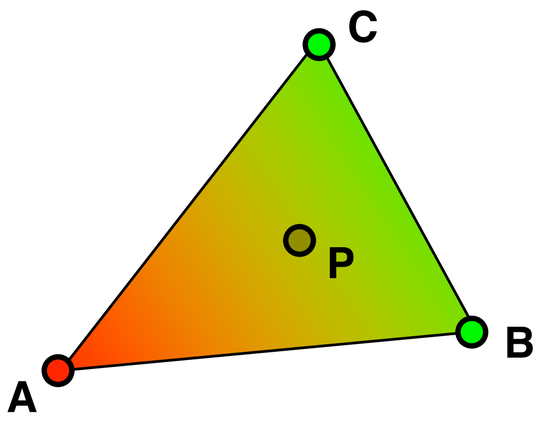
Barycentric Interpolation

- Barycentric coordinates:
  - $\mathbf{p} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}$ with $\alpha + \beta + \gamma = 1$
  - Unique for non-collinear **a**,**b**,**c**
  - Ratio of triangle areas
  - $\alpha(\mathbf{p})$, $\beta(\mathbf{p})$, $\gamma(\mathbf{p})$ are linear functions
  - Gives inside/outside information



Barycentric Interpolation

- Barycentric coordinates:
  - $\mathbf{p} = \alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c}$ with $\alpha + \beta + \gamma = 1$
  - Unique for non-collinear **a**,**b**,**c**
  - Ratio of triangle areas
  - $\alpha(\mathbf{p})$, $\beta(\mathbf{p})$, $\gamma(\mathbf{p})$ are linear functions
  - Gives inside/outside information
  - Use barycentric coordinates to interpolate vertex normals (or other data, e.g. colors)

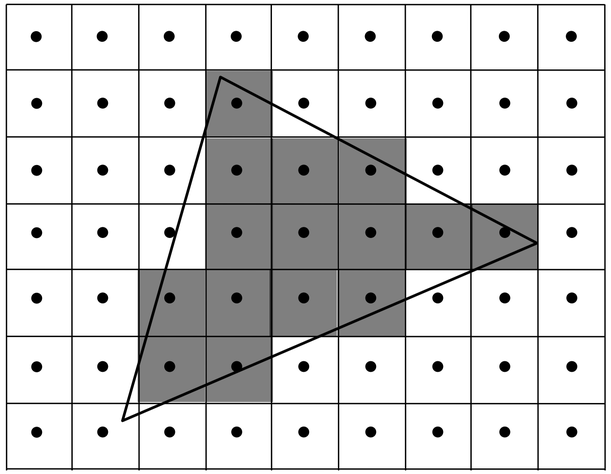


$$\mathbf{n}(\mathbf{P}) \quad = \quad \alpha \cdot \mathbf{n}(\mathbf{A}) + \beta \cdot \mathbf{n}(\mathbf{B}) + \gamma \cdot \mathbf{n}(\mathbf{C})$$

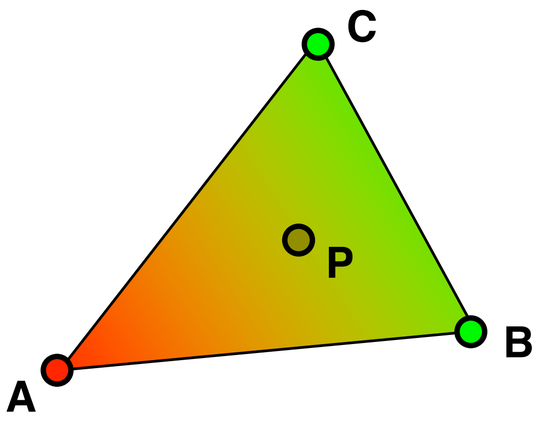
Triangle Rasterization

- Each triangle is represented as three 2D points (x_0, y_0), (x_1, y_1), (x_2, y_2)
- Rasterization using barycentric coordinates

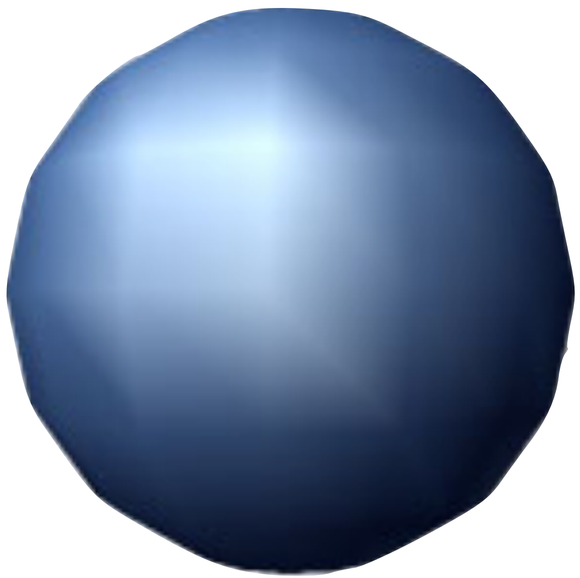
for all y **do**
  for all x **do**
    compute (α,β,γ) for (x,y)
    if ($\alpha \in [0,1]$ and $\beta \in [0,1]$ and $\gamma \in [0,1]$
      set_pixel (x,y)



Color Interpolation

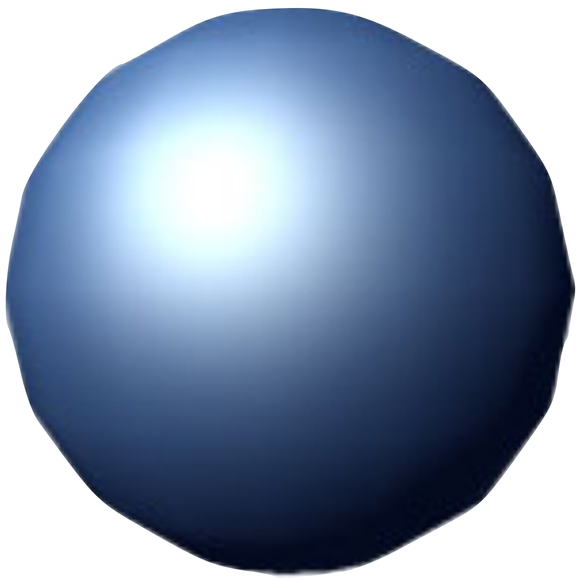


Per-vertex



Evaluate color on vertices, then interpolates it

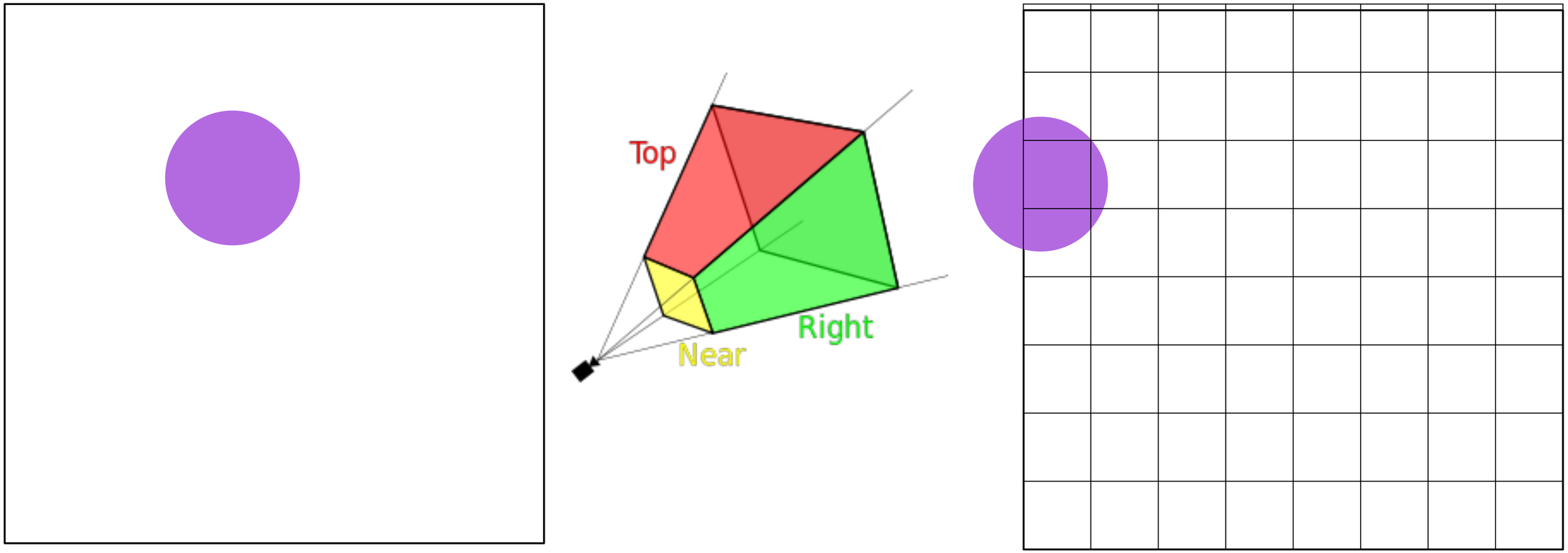
Per-pixel



Interpolates positions and normals, then evaluate color on each pixel

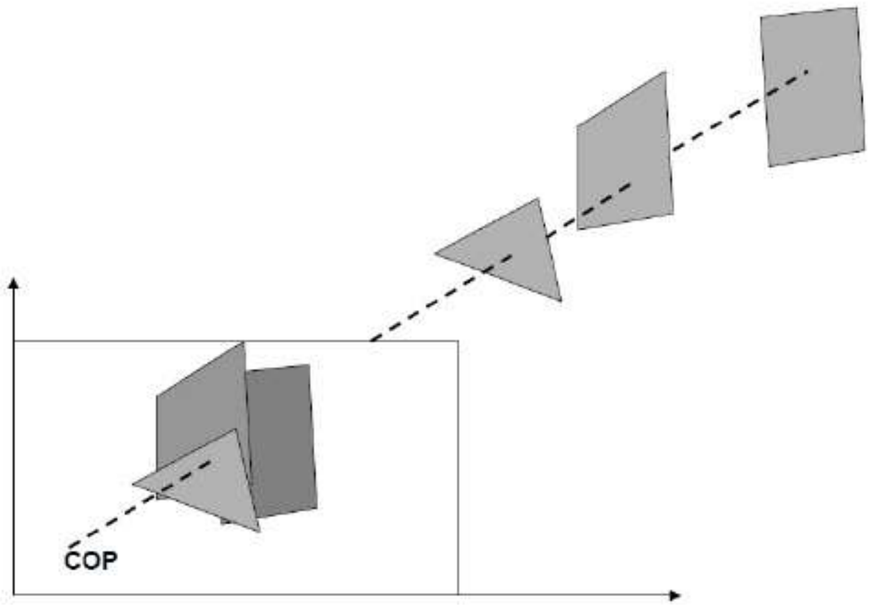
http://www.cgchannel.com/2010/11/cg-science-for-artists-part-2-the-real-time-rendering-pipeline/

Clipping



Objects Depth Sorting

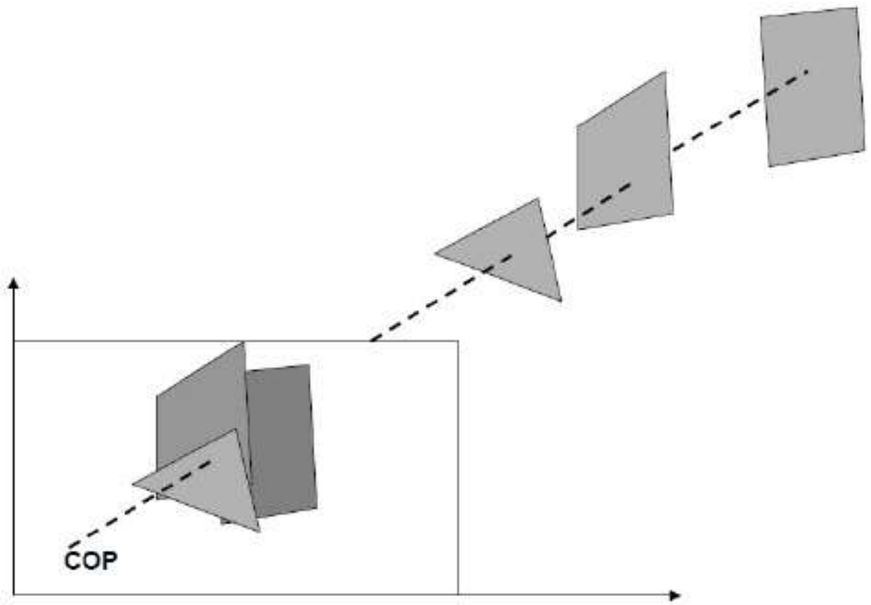
- To handle occlusion, you can sort all the objects in a scene by depth

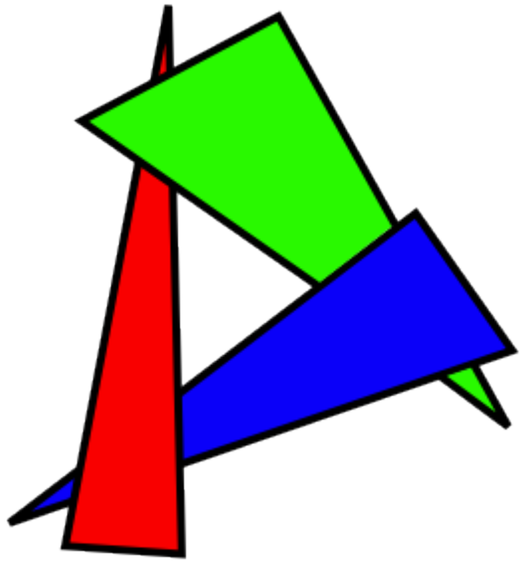


Objects Depth Sorting

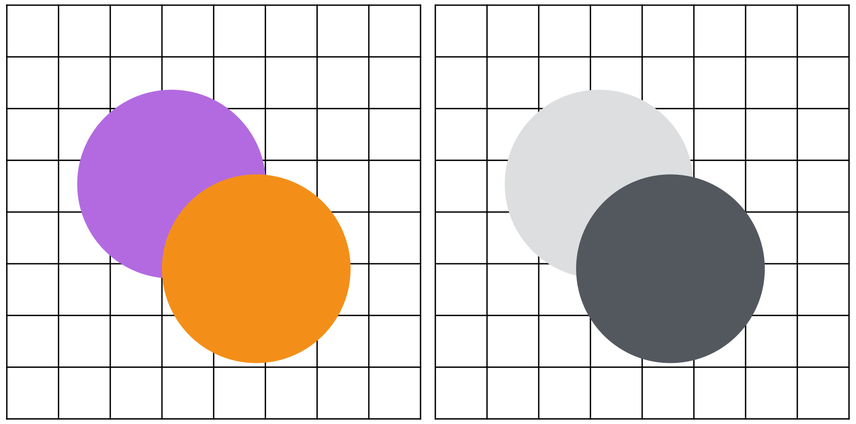
- To handle occlusion, you can sort all the objects in a scene by depth

- This is not always possible!





z-buffering

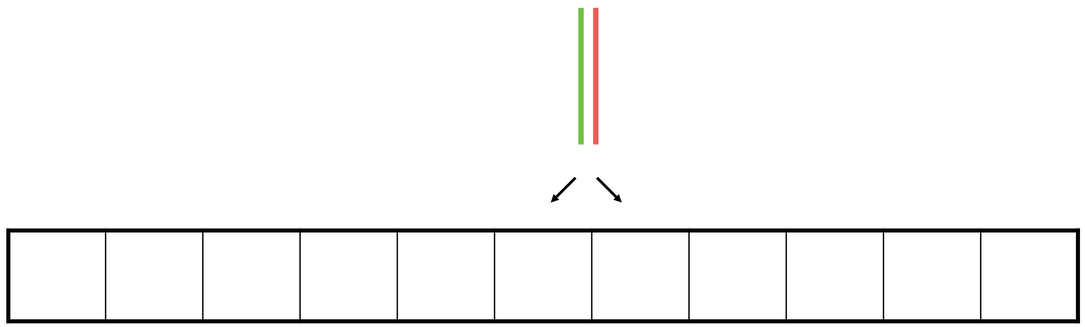


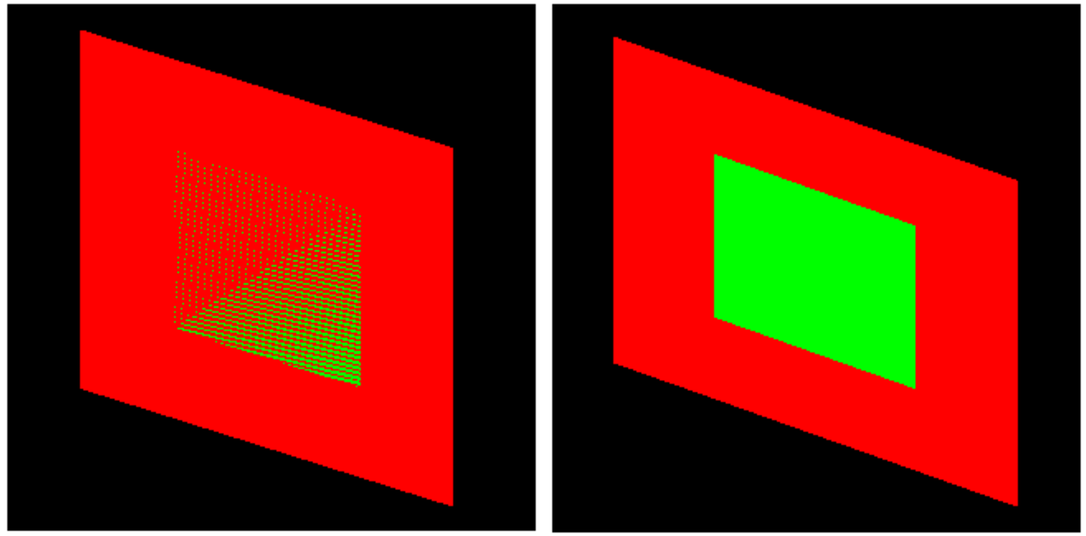
- You render the image both in the Image and in the depth buffer, where you store only the depth
- When a new fragment comes in, you draw it in the image only if it is closer
- This always work and it is cheap to evaluate! It is the default in all graphics hardware
- You still have to sort for transparency…

z-buffer quantization and “z-fighting”

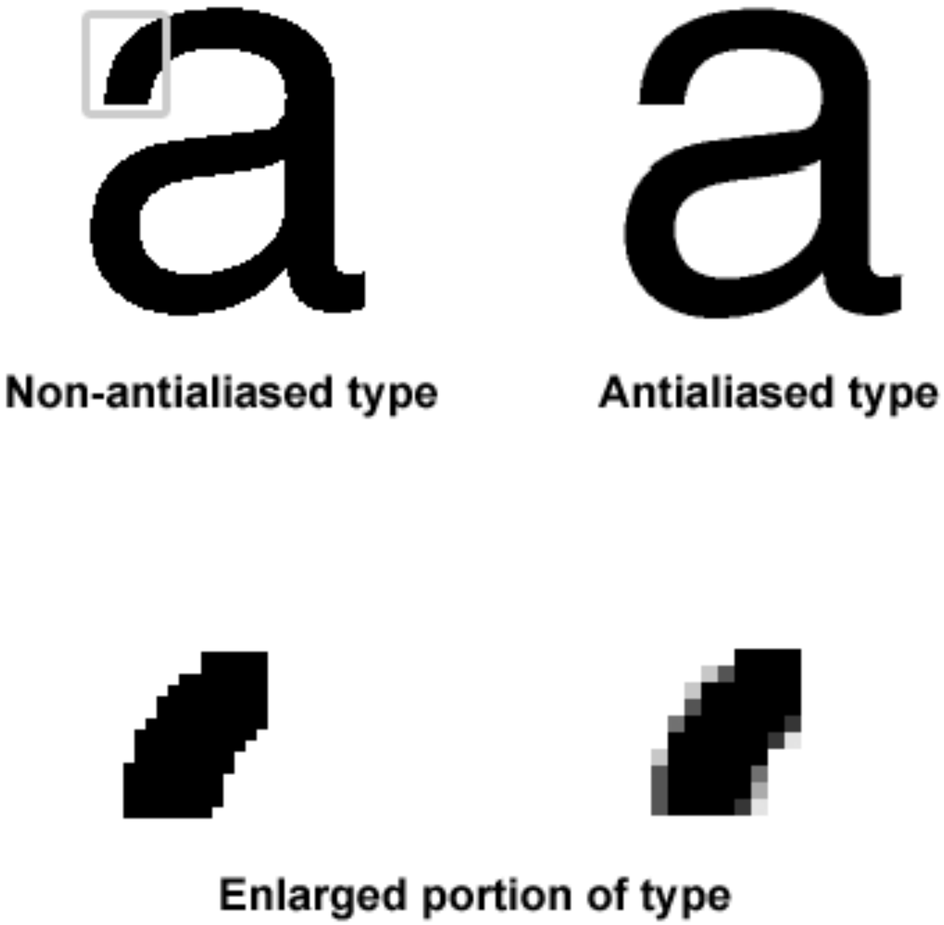
- The z-buffer is quantized (the number of bits is heavily dependent on the hardware platform)

- Two close object might be quantized differently, leading to strange artifacts, usually called “z-fighting”





Super Sampling Anti-Aliasing



- Render nxn pixels instead of one
- Assign the average to the pixel

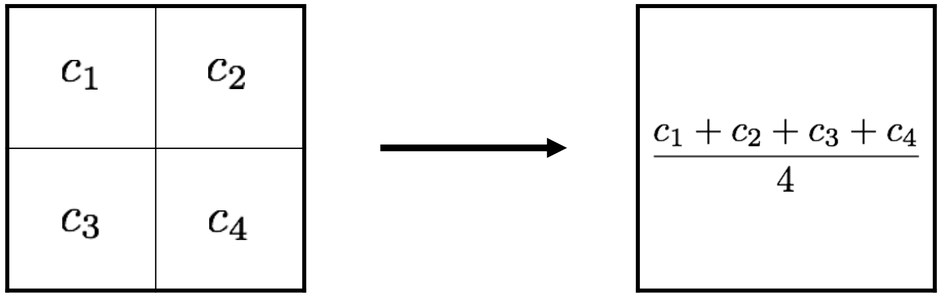


Image Copyright: Fritz Kessler

References

Fundamentals of Computer Graphics, Fourth Edition
4th Edition **by Steve Marschner, Peter Shirley**

Chapter 8